

# Jay Liu

## Senior Software Developer

jayi.nerd@gmail.com | Toronto, Canada | linkedin.com/in/jay-liu-dev | j-liu-sys.github.io

### Professional Experience

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#### Software Developer

09/2022 – Present | Toronto, Canada

##### Arctic Wolf

- Developed cloud-native backend services in **Go and Python** powering Arctic Wolf's cybersecurity platform, processing **billions of security events daily** to support threat detection, investigation, and response workflows.
- Built distributed streaming and correlation pipelines using **Kafka, AWS, Kubernetes, Redis, and Docker** to ingest, enrich, and analyze large-scale security telemetry from customer environments.
- Designed and implemented microservices responsible for transforming raw security data into actionable threat intelligence, enabling security analysts to identify and respond to threats more efficiently.
- Developed scalable APIs and event-driven services supporting real-time security analytics, contextual enrichment, and automated investigation workflows.
- Optimized high-throughput distributed systems handling massive data volumes, improving service reliability, scalability, and processing performance across production environments.
- Built Python-based data processing services that standardized and enriched security datasets for downstream analytics, machine learning, and AI-assisted security operations use cases.
- Partnered with security researchers, analytics teams, and platform engineers to deliver backend infrastructure supporting advanced threat detection and intelligent security automation initiatives.
- Enhanced observability through monitoring, logging, and alerting improvements, increasing operational visibility and reliability for mission-critical cybersecurity services.
- Automated CI/CD and deployment workflows using Jenkins, Docker, and Kubernetes, accelerating software delivery while maintaining platform stability and security.

**Technologies:** Go, Python, AWS, Kafka, Kubernetes, Docker, Redis, Jenkins, Microservices, Distributed Systems, REST APIs

#### Software Developer

06/2022 – 08/2022 | Toronto, Canada

##### Hootsuite

- Developed backend microservices supporting identity, authentication, authorization, and compliance capabilities for millions of users across Hootsuite's social media management platform.
- Designed and maintained scalable APIs and distributed services responsible for user identity management, access control, account lifecycle management, and platform security.
- Built cloud-native services on AWS and Kubernetes supporting enterprise-grade authentication and authorization workflows across multi-tenant SaaS environments.
- Contributed to automated compliance and governance solutions that ensured secure handling of customer identity and account data.
- Improved service scalability, reliability, and operational efficiency through infrastructure optimization, deployment automation, and performance enhancements.
- Collaborated with platform engineering teams to deliver secure backend services supporting enterprise customer onboarding, account administration, and user management experiences.
- Enhanced observability and production monitoring capabilities to improve incident response and operational troubleshooting.
- Participated in architectural discussions focused on building extensible platform services capable of supporting intelligent automation and future AI-driven workflow enhancements.

**Technologies:** Go, AWS, Kubernetes, Docker, REST APIs, IAM, Authentication, Authorization, Microservices, Jenkins

#### Software Developer

07/2021 – 06/2022

##### Citi

- Developed distributed backend applications supporting enterprise risk data management and analytics platforms used by internal trading and risk management teams.
- Contributed to a centralized risk reference data platform that improved data consistency, accessibility, and decision-making across multiple business lines.
- Designed and implemented automated testing frameworks for high-performance microservices, reducing validation effort by over **90%** and significantly accelerating release cycles.
- Expanded risk reporting capabilities by integrating additional reference data sources, enabling more accurate counterparty credit risk calculations across five lines of business.
- Built backend services and data processing workflows that transformed complex financial datasets into standardized information consumable by downstream analytics and reporting systems.
- Developed APIs and integration services supporting large-scale risk data aggregation, validation, and distribution across enterprise applications.
- Collaborated with engineering and business stakeholders to deliver scalable solutions supporting trading, regulatory, and risk management requirements.
- Optimized CI/CD pipelines and automation tooling following platform modernization initiatives, improving software quality, deployment efficiency, and operational reliability.

**Technologies:** Python, Node.js, Java, JavaScript, MongoDB, REST APIs, Microservices, Distributed Systems, CI/CD

### Education

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#### Bachelor of Science

09/2016 – 07/2021 | Alberta, Canada

University of Alberta